

Atomic Emission

Table of Contents

- [Learning Objectives](#)
- [Suggested Reading](#)
- [Experiment Description](#)
- [Lab To Do Check List](#)
- [Lab Grade Breakdown](#)
- [Background Information](#)
- [Fun Facts About Atomic Emission](#)
- [Apparatus and Chemicals](#)
- [Lab Procedure](#)
- [Extra Problems](#)

Learning Objectives

After having completed this lab experiment, students will

Know:

- How atomic emission can be measured
- That light emission spectra can be linear and continuous

Understand:

- That atomic emission is observed when the excited atom electrons move back from a higher energy state to a lower energy state
- The difference between various light sources such as incandescent, LED and gas discharge lamps
- That electromagnetic radiation can have different frequencies/energy and that the atomic emission spectra will reflect the frequencies of the emitted photons when electrons transition from higher to lower energy states
- That each atom produces a unique atomic emission spectrum

Do:

- Observe emission spectra directly by looking through a diffraction grating
- Use a spectrophotometer to measure and record emission from different light sources in a visible light range
- Measure and quantify the atomic emission of hydrogen
- Measure the atomic flame emission of a series of metal salts

Suggested Reading

[Chemistry: Atoms First](#) from OpenStax, Print ISBN 1947172646, Digital ISBN 1947172638, chapter 3.1 and 3.2.

Experiment Description

In this lab you will perform 4 mini-experiments. The order in which you perform the experiments does not matter. You will move between the 4 stations set up in the lab and complete each mini experiment in a group of 2.

1. You will measure and quantify the emission spectrum from a hydrogen discharge lamp.
2. You will measure the emission spectrum a series of gases in provided discharge lamps.
3. You will use a flame to excite metallic salts and observe the emission spectra of these metals.
4. You will investigate the spectra of a number of commercial light sources, such as incandescent lamp, white LED and a fluorescent day light lamp.

Lab To Do Check List

TO DO before coming to the lab:

- Read the lab manual
- Do the pre-lab quiz to gain access to the datasheet and the lab report
- Print the Data Sheet/Results
- Prepare a procedure flowchart, print it and bring it with you.

TO DO during the lab:

- Get your flowchart reviewed and signed by the TA
- Complete all five mini-experiments in groups of 2
- Fill out your Data Sheet/Results **INDIVIDUALLY** and get it signed by your TA.
- Clean up

TO DO after the lab:

- Convert your Data Sheet/Results and Procedure Flowchart into a **SINGLE** pdf file and submit on MyCourses
- Each student submits their work **INDIVIDUALLY**.

Lab Grade Breakdown

This lab grade is 25 points total. The breakdown for this lab is as follows.

Lab	Points	Weight, % from the total lab grade	Due date
Lab 2: Atomic Emission	25	12.5%	
Pre-Lab Quiz	5		Before the lab
Procedure flowchart	Feedback only		Before the lab
Data Sheet/Results	20		Completed by the end of the lab. Submitted by the end of the lab day.

Background Information

When a substance absorbs energy, electrons are excited to higher potential energy levels/states further away from the nucleus. This excitation could be achieved through heating, applying voltage or electromagnetic radiation to the substance. The electrons will eventually return to their ground state, releasing the absorbed energy as electromagnetic radiation, which is called emission. The energy of the released photons will be equal to the energy between two states. In this experiment, we will be investigating the radiation that is emitted in the visible to the human eye region of the electromagnetic spectrum, the region between 400 – 750 nm.

The emitted radiation can be collected and analyzed by a spectrophotometer that will generate an emission spectrum. Emission spectra can be continuous or linear depending on the type of emission. Examples of continuous and linear spectra are shown on Figures 1-3.

Substances will emit light when they are heated to a high enough temperature (black body radiation). When a solid like a fireplace poker (normally made of cast iron) is heated sufficiently, it will emit red light (Fig. 1, A)). This light source is continuous, meaning it emits a broad range or an unbroken series of wavelengths and produces *continuous* emission spectrum.

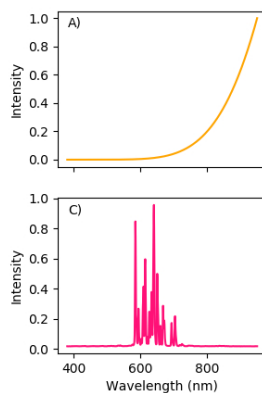


Figure 1 -) Computer generated (simulated) visible spectrum of a hot fireplace poker. C) Emission Spectrum of the Neon (Ne) discharge lamp.

Similarly, heating a tungsten filament in an incandescent bulb will produce a continuous emission spectrum (Fig. 2). The light spectra emitted from the sun, or the stars will also be continuous (Fig. 3, B).

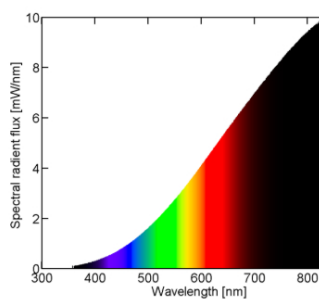


Figure 2 - Spectral power distribution of a 25 W incandescent light bulb (the image is used under a free license from https://en.wikipedia.org/wiki/Incandescent_light_bulb)

Radiation emitted by excited individual atoms is not continuous: it is discrete and produces *Line* spectra. Thus, in our computer program, we will see several discrete lines (Fig. 4

spectra. Thus, in our computer readout from our Spectrometer, we will see several discrete lines (Fig. 1, C).

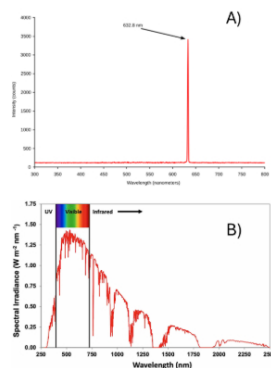


Figure 3 - A) Experimentally obtained spectrum of a red laser pointer (the image is used under a free license from https://commons.wikimedia.org/wiki/File:Helium_neon_laser_spectrum.svg) B) Standard Solar Radiation Spectrum at Sea Level (based upon National Renewable Energy Laboratory data) <https://www.nrel.gov/grid/solar-resource/spectra.html>

The observation that excited atoms generate line spectra of fixed wavelengths indicates that the atom can undergo discrete energy changes. These line spectra are regular and fixed. We can thus use line spectra as a "fingerprint" for the atom. The emission spectrum is unique for each atom in the same way that fingerprints are unique for each person. In this way we can use emission spectra as a means of identifying the constituent atoms of a sample.

You already utilized this property of each atom to emit a unique spectrum during the Sherlock Holmes mystery experiment, where you determined the identity of metal atoms in the salts of Na, K and Mg by subjecting them to a flame test. In this experiment you will be repeating the flame test and, this time, you will be also recording the emission spectra using a spectrophotometer and an optic probe and identifying unknown metal salts with this method.

Monochromatic light sources like laser pointers emit a single wavelength of radiation and produce monochromatic (only one line is present) spectrum (Fig 3, A)).

The discharge lamps generate light when the voltage is applied to the ionized gas (plasma) they contain. Most of the time, the discharge lamps carry inert gases such as argon (Ar), neon (Ne), krypton (Kr) and xenon (Xe). The light is generated as a result of atomic emission from the atoms of these elements and the emission spectra are linear. They have multiple narrow line widths in the visible spectral region being investigated (see Fig. 1, C) for the Ne discharge lamp spectrum).

In this experiment you will also experiment with emissions from the discharge lamps containing hydrogen gas (H_2), carbon dioxide (CO_2) and nitrogen (N_2). As you move from monoatomic noble gases to multiatomic molecules, the spectra become more complicated in appearance and the spectral lines will overlap to give broader bands.

In hydrogen discharge tube, hydrogen gas is in its diatomic state and it splits into two hydrogen atoms as the a high voltage is applied to the tube.

The commercial LED lights generate light when the current passes through a semiconductor material, where electrons combine with the "electron holes" (positively charged spaces where electrons could exist) and generate photons/light as a result. This type of emission is called luminescence and it generates "cold light" as opposed to incandescence, where the light emits when an object is heated to high temperatures. Common semiconductors used in LEDs are so called III-V materials made of group III and V elements, such as aluminum gallium arsenide, gallium phosphide, etc. The white LED lamps mimic incandescent bulbs through application of layered semiconductor materials, each of which produces emission at a specific wavelength range and all together they add up to give a continuous emission spectrum.

How can we observe a light spectrum?

Electromagnetic radiation is characterized by its wavelength λ and frequency ν . For any wave, the product of the wavelength and frequency of the wave is equal to its speed (Eq. 1). For electromagnetic radiation travelling in a vacuum, this speed is the speed of light (c): $2.9978 \times 10^8 \text{ m s}^{-1}$.

$$\nu\lambda = c \quad (1)$$

Where:

- λ : the wavelength (m)
- c : the speed of light in vacuum ($2.9978 \times 10^8 \text{ m s}^{-1}$)
- ν : the frequency (s^{-1} or Hz)

The wavelength of visible light is normally expressed in nanometers (nm). Thus, when conducting calculations, you first need to convert the *nm* to *m* by multiplying the value in *nm* by 10^{-9} .

Example:

What frequency corresponds to a wavelength of light of 450.0 nm?

Solution

$$v = \frac{c}{\lambda} = \left(\frac{2.9978 \times 10^8 \text{ m/s}}{450.0 \text{ nm} \times 10^{-9}} \right) = \left(\frac{2.9978 \times 10^8 \text{ m/s}}{450.0 \text{ nm}} \right) \left(\frac{10^9 \text{ nm}}{1 \text{ m}} \right) = 6.662 \times 10^{14} \text{ s}^{-1}$$

As discussed above, most of the light sources emit radiation at multiple wavelengths. In order to identify and record all the different wavelengths, we need to separate light into its individual components.

When you are looking directly at the light produced from the atomic emission in the visible spectrum range, you will see a color. Your eye will not be able to split this color into its individual components.

One way to separate light into individual components is to use a prism. For example, if we pass the white light from the sun through a prism, we will see distinct spectral colors. Every component of the continuous light source is bent (refracted) by the prism to a different extent, forming a continuous spectrum. One example we see in everyday life is when sunlight is dispersed by a raindrop: the raindrops can act like a prism, forming a continuous spectrum that we observe as a rainbow.

We can also use a diffraction grating to separate color into its component. In our experiment, we will look at the light directly through a diffraction grating and also use a spectrophotometer to observe and record spectra from different light sources. A spectrophotometer is equipped with its own diffraction [grating](#) instead of a prism to improve the resolution of the spectrum and reduce the cost and size of the instrument. Our spectrophotometers also have a narrow slit between the radiation source and the diffraction grating which improves the resolution even further. The spectrophotometer then collects the signal on a CCD (charge-coupled device) Array, which is very similar to the sensor on a digital camera (Fig. 4).



spectrometer

Figure 4 - Schematic representation of how the light travels from a light source to a detector in a spectrophotometer

This CCD Array allows us to digitally capture the spectrum. On the computer, we will see our spectrum represented as a graph, with the wavelength of light on the x-axis and the intensity of the signal on the y-axis (Fig. 5)

Figure 5 - An example of a spectral analysis software screen

Hydrogen atomic emission spectrum.

The Balmer series.

The emission spectrum of hydrogen atom consists of so called spectral series, each series is defined by the quantum number of the lowest energy orbital to which transition occurs (Fig. 6). For example, Lyman series correspond to the electron transitions from $n_i > 1$ orbitals to the $n_f = 1$ orbital.

In this experiment we will be recording the hydrogen emission spectrum in the visible region. The four electron transitions responsible for the emission in the visible region belong to the Balmer series and correspond to transitions from the orbitals with $n_i \geq 2$ to the $n_f = 2$ orbital.

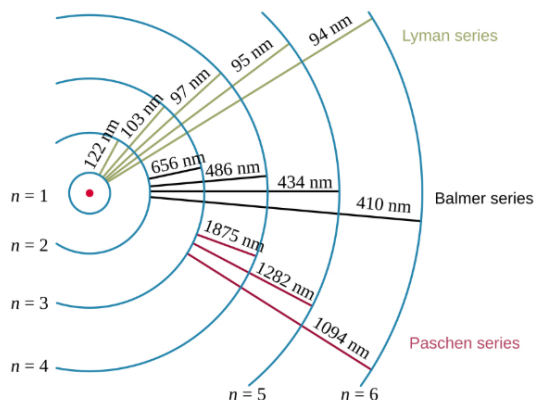


Figure 6 - Electron transitions for the first three hydrogen atom atomic emission series and their corresponding wavelengths. (the image is used under a free license from A_hydrogen_szinkepi.jpg; User:Szidoridervative work: OrangeDog (talk • contribs), CC BY 2.5 <<https://creativecommons.org/licenses/by/2.5/>>, via Wikimedia Commons)

Johann Jacob Balmer (1825-1898) was a math teacher at a girl's high-school in Basel. Balmer became interested in four hydrogen visible spectral lines, 656 nm, 486 nm, 434 nm, 410 nm that were measured by Ångström in 1871. By then, the atomic emission spectra for many metals had been also observed and recorded and scientist were trying to find an explanation to where these lines came from. Balmer, however, was not so much interested in the physical phenomenon behind the lines, but in numbers per se. He wanted to find a pattern, a mathematical formula that would allow not only to calculate the four

lines, but other lines that have not been recorded experimentally yet. In 1885, Balmer published his formula connecting the wavelengths of the spectral lines to the integer numbers.

$$\lambda = \text{const} \left(\frac{m^2}{m^2 - n^2} \right) \text{ where } n = 2$$

In 1890 Rydberg generalized Balmer's formula as

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where $R_H = 1.097 \times 10^7 \text{ m}^{-1}$ and it is called **Rydberg constant**.

Bohr model.

According to Bohr model, each orbital has a fixed energy level and, therefore, the energy difference ($\Delta E = E_f - E_i$) between any two orbitals is a constant. The photon emitted as a result of transition between two specific orbitals has always the same wavelength.

The energy of an emitted photon reflects the difference between the energies of the final and initial electron energy levels and is defined as (Eq. 2):

$$\Delta E = h\nu \quad (2)$$

where:

- E : the energy of the photon (J)
- h : Planck's constant ($6.626 \times 10^{-34} \text{ J}\cdot\text{s}$)
- ν : the frequency (s^{-1} or Hz)

Additionally due to the relationship between frequency and wavelength in Equation 1.

$$\Delta E = \frac{hc}{\lambda} \quad (3)$$

The emission always occurs from a higher energy level (E_i) to a lower energy level (E_f). Therefore, $\Delta E = E_f - E_i$ for emission is < 0 .

We can use the Bohr equation to relate the ΔE of the observed emission to the principal quantum numbers of the energy state from and to which the transition occurred.

Therefore, In order to determine the principal quantum numbers for the hydrogen transition, we will use the following equation:

$$\Delta E = -B_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \quad (1)$$

where:

- ΔE : energy of the transition (J)
- B_H : Bohr constant ($2.18 \times 10^{-18} \text{ J}$),
 $B_H = hcR_H$, where $R_H = 1.09678 \times 10^7 \text{ m}^{-1}$ (Rydberg constant)
- n_f : principal quantum number of the energy level to which transition has occurred (no units)
- n_i : principal quantum number of the energy level from which transition has occurred (no units)
- when $n_f < n_i$, the emission is observed ($\Delta E < 0$)
- when $n_f > n_i$, the absorption is observed ($\Delta E > 0$)

Example:

Calculate the energy of the photon emitted when an electron in a hydrogen atom moves from $n=3$ to $n=1$.



Solution

$$\begin{aligned} \Delta E &= B_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \\ \Delta E &= -B_H \left(\frac{1}{1^2} - \frac{1}{3^2} \right) \\ \Delta E &= -2.18 \times 10^{-18} \times \left(\frac{8}{9} \right) = -1.94 \times 10^{-18} \text{ J}, \quad |\Delta E| = 1.94 \times 10^{-18} \text{ J} \end{aligned}$$

Flame Atomic Emission Spectroscopy.

Flame Atomic Emission Spectroscopy (FAES) is a type of atomic emission spectroscopy used to detect and analyze the presence of certain metal ions by using flame as a source of thermal energy to excite electrons. In fact, you will use FAES in your next laboratory experiment, Mystery, to determine which metal ion is present in your unknown white powder.

When the metal salt sample is introduced into the flame, the flame might take on a characteristic color depending on the metal present.

How does it work? In the flame, the metal salt evaporates, dissociates under continuous heating and becomes a gas mixture of individual atoms. The thermal energy of the flame also excites the electrons in these atoms into their corresponding excited electronic states. When these electrons return to their ground state energies, they emit light of wavelengths characteristic to their metal, allowing us to

determine which metals are present in the solution. The flame colors for some metal salts are shown in Figure 7.

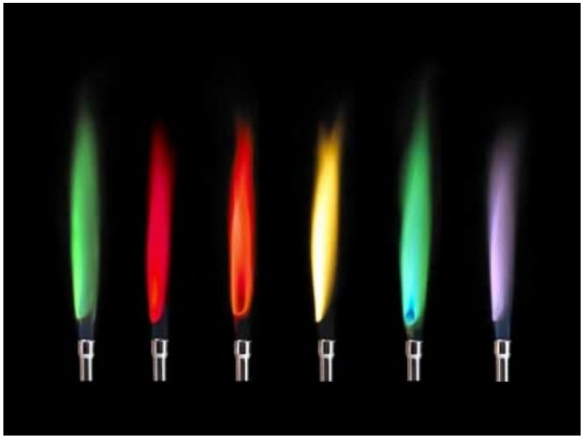


Figure 7 - Flame test results: zinc, lithium, strontium, sodium, copper, potassium

Fun Facts About Atomic Emission

Fireworks

Perhaps, one of the most enjoyable manifestations of the atomic emission phenomenon are the fireworks colors. As early as 200 BC Chinese pyrotechnics discovered how to color their fireworks. They would add metal salts naturally occurring in minerals to the explosive content of a firework to produce colored flames in the sky. Look at the firework below. Which metals, do you think, are responsible for the pink-red and green firework?



Figure 8 - Fireworks in the Old Port by the Clock Tower, Montreal (the image is used under a free license from Michael Vesia, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons, https://commons.wikimedia.org/wiki/File:Montreal_Clock_Tower_Fireworks.jpg)

Hydrogen-alpha

The red (656 nm) atomic hydrogen emission spectral line (Balmer series) is called Hydrogen-alpha line. This line is very useful to astronomers who observe it when studying composition of emission nebulae as well as Sun atmosphere, which consist primarily of hydrogen. The map below shows a distribution of ionized hydrogen in the Milky way as observed from the Northern Hemisphere using a hydrogen-alpha Mapper.

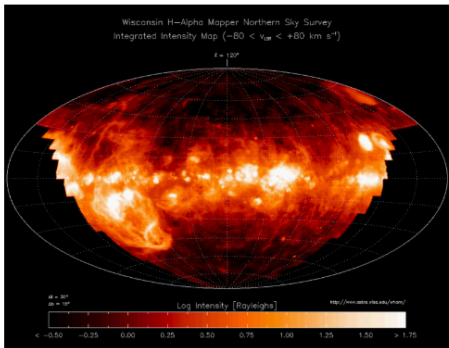


Figure 9 - Ionized gas in the Milky Way Galaxy, observed by the <http://www.astro.wisc.edu/wham/> Wisconsin H-Alpha Mapper (WHAM) Northern Sky Survey (Haffner, L. M. et al, 2003, Astrophysical Journal Supplement, 149(2), 405). The Wisconsin H-Alpha Mapper is funded by NSF

Apparatus and Chemicals

Apparatus

Item	Quantity
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Spectrometer	1
Fiber optic cable	1
Discharge Lamps	6
Bunsen Burner	1
Commercial lamps	5

Chemicals

Chemical
Potassium Chloride
Sodium Chloride
Lithium Chloride
Strontium Chloride

Chemical Safety and Safety Data Sheets (SDS)

Please review the SDS (click on the hyperlink to access SDS directly) for each chemical you will be using in the lab.

- [Sodium chloride](#) (table salt): Slightly hazardous in case of skin contact (irritant), of eye contact (irritant), of ingestion, of inhalation.
- [Potassium chloride](#): Slightly hazardous in case of skin contact (irritant), of eye contact (irritant), of ingestion, of inhalation.
- [Lithium chloride](#): Causes eye and skin irritation. Harmful if swallowed or absorbed through the skin. May cause central nervous system effects.
- [Strontium \(II\) chloride](#): May cause eye, skin, and respiratory tract irritation. The toxicological properties of this material have not been fully investigated.

Lab Procedure

Print Procedure

Experiment I - Observation of light emission from Hydrogen gas discharge lamp

Figure 10 -

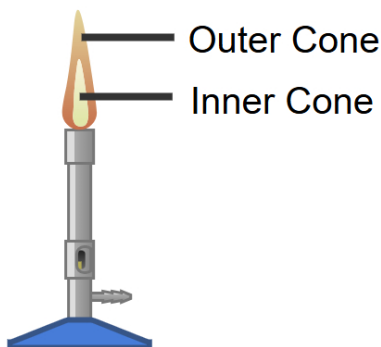
1. Click on "Untitled" on the top left of the program (Fig. 10, A) and then select "New Experiment". The program will ask you if you want to save changes. Select "Don't Save".
2. A menu should appear. Select "Emissions" from the menu and then select "vs. Wavelength (Full Spectrum)". You are now ready to take your first scan.
3. Turn on the lamp using the switch at the back.
4. The emission fiber optic probe is already set up and secured in front of the lamp.
5. Record the color of light (visual observation) in your datasheet.
6. Now look at the light through the diffraction grating card and observe individual spectral lines. Write down the color of each line.
7. To obtain the actual wavelength values for the lines you are seeing, you need to use a spectrometer.
8. Press the "Collect" button (Fig. 10, B) on the software window to begin data collection.
9. Once you see a spectrum appear on the screen press the "Stop" button.
10. You should see two clear peaks on the spectrum. If either of the peaks has a flat top, it means the detector is saturated (it receives more light than it can measure). In this case you should do the following: on the right of the screen, you should see a data table. Next to "Data Set 1" (Fig. 10, C), click on the three dots and click on "Delete Data Set". Move the probe slightly further away from the lamp and press collect again. Your goal is to make the two intense peaks on the spectrum become sharp.
11. For each peak, move your cursor to the top of the peak. Click the left trackpad button. You should see the wavelength (Fig. 10, D) corresponding to the peak, at the bottom of the screen. Record the wavelength (nm units) for each peak. You might see another small peak on the spectrum, ignore it for now, we will look at it in the next step.
12. To determine the wavelengths of the less intense peaks, you will need to increase the signal to the detector. To do this, follow steps 13 and 14.
13. Move the fiber optic closer to the lamp. Get as close as you can without the fiber optic touching the lamp.
14. Increase the integration time by clicking on the gear in the top right of the software. Change the integration time from 30s to 100s.
15. Press the "Collect" button on the software to begin data collection.
16. Once you see a spectrum appear on the screen press the "Stop" button.
17. Click on the Y-axis label "Intensity" (Fig. 10, F) on the left side of the screen. A small screen will appear. Deselect "Intensity" for Data Set 1. This will clear the first run and will let us analyze the second run. Check if "Intensity" is selected for Data Set 2. If it is not, you should select it.
18. You should now see two peaks below 450 nm. They will be much smaller than the more intense peaks, but you should still see it to be able to see them and to determine its wavelength. Record the wavelength of the two small peaks you see between 400 and 450 nm onto your datasheet.
19. Turn off the lamp and move the clamp back to place it at about 5 cm from the lamp.
20. Fill the table for this station in your Data Sheet/Results and do the calculations.

Experiment II - Observation of light emission from flames.

In part II you will collect and record the flame emission spectra for a series of metal chloride salts.

1. Click on "Untitled" on the top left of the program (Fig. 10, A) and then select "New Experiment". The program will ask you if you want to save changes. Select "Don't Save".

2. A menu should appear. Select "Emissions" from the menu and then select "vs. Wavelength (Full Spectrum)". You are now ready to take your first scan.
3. The setup at your station should contain the following. A Bunsen burner and 4 vials containing LiCl, NaCl, KCl, and SrCl₂.
4. The setup is prepared for you, please do not touch the setup with the flame and fiber optic. If there is a problem with the setup, please let your TA know. DO NOT try to adjust yourself.
5. For this next step one student should be dealing with the Bunsen burner and a second student should be operating the computer. As a group you should switch roles often so you each get to practice with the different roles.
6. Your TA will help you turn on the Bunsen burner.
7. Take the Nichrome wire connected to the NaCl bottle. Dip the tip of the wire into the small beaker of water labelled (NaCl). It is very important that we don't cross contaminate our samples. Use only the NaCl wire and NaCl water beaker when dealing with the NaCl sample.
8. Dip the moistened wire into the NaCl vial. Some salt should stick to the tip of the wire.
9. Press the (Fig. 10, B) "Collect" (Fig. 10, B) button on the software to begin data collection.
10. Carefully place the tip of the nichrome wire containing the NaCl salt into the Bunsen Burner. You want to place the wire into the tip of the inner cone of the flame.

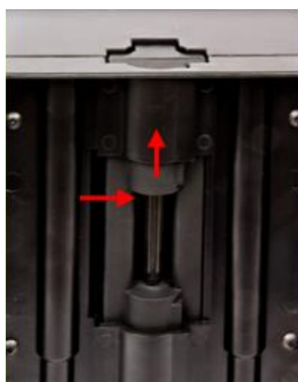


11. Once you see a spectrum appear on the screen press the "Stop" button.
12. Record the color of light by visual observation in your datasheet.
13. On the right of the screen, you should see a data table. Next to (Fig. 10, C) "Data Set 1" (Fig. 10, C), click on the three dots and click on "Rename Data Set". Rename the data set based on the salt that you are using. For example, name the dataset "NaCl" if you are using NaCl.
14. If there is more than one spectrum displayed on the graph (this will be the case once you scan your second salt) you can focus on the most recent scan by clicking on the Y-axis label (Fig. 10, E) "Intensity" on the left side of the screen. Deselect "Intensity" for any but the most recent spectra. This will clear the data to allow us to prepare for the next scan.
15. Record the wavelengths of the five most intense peaks (if your spectrum has less than five peaks, list all the peaks) on the spectra. For each peak, move your cursor to the top of the peak. Click the left trackpad button. You should see the wavelength corresponding to the peak at the bottom of the screen. Make sure to mark down the wavelength (nm units).
16. Repeat steps 7 - 15 and record the emission spectra of all 4 salts on your Data Sheet/Results.

Experiment III - Observation of light emission from elemental gas discharge lamps.

In experiment III, you will record the emission from a series of gas discharge lamps.

1. The atomic emission lamp at this lab station has the following lamps: nitrogen, carbon dioxide, helium, argon, hydrogen, neon. There is a fiber optic cable aimed at the lamp. This fiber optic cable is also attached to the spectrophotometer.
2. Click on "Untitled" on the top left of the program (Fig 10, A) and then select "New Experiment". The program will ask you if you want to save changes. Select "Don't Save".
3. A menu should appear. Select "Emissions" from the menu and then select "vs. Wavelength (Full Spectrum)". You are now ready to take your first scan.
4. Make sure the lamp is turned off. It is very important to turn off the lamp whenever you change bulbs.
5. If there is a bulb in the lamp remove it by placing your finger under the top ledge of the illuminated area and lifting the bulb until you can grab it from the top. (see below)



6. Take one of the discharge lamps and insert it into the base. The lamps are designed so that they can only be inserted in the correct orientation. Once correctly inserted into the opening gently push down from the top until the bulb is well seated at the bottom of the well. Turn on the lamp.
7. Record the color of light by visual observation in your datasheet.

8. Now look at the light through the diffraction grating card and observe individual spectral lines. Write down the color of each line.
9. To obtain the actual wavelength values for the lines you are seeing, you need to use a spectrometer.
10. Press the (Fig. 10, B) "Collect" button on the software to begin data collection.
11. Once you see a spectrum appear on the screen press the "Stop" button.
12. On the right of the screen, you should see a data table. Next to (Fig. 10, C) "Data Set 1", click on the three dots and click on "Rename Data Set". Rename the data set based on the lamp that you are using. For example, name the dataset "Air" if you are using the air lamp.
13. If there is more than one spectrum displayed on the graph (this will be the case once you scan your second gas) you can focus on the most recent scan by clicking on the Y-axis label (Fig.10, E) "Intensity" on the left side of the screen. Deselect "Intensity" for any but the most recent spectra. This will clear the data to allow us to prepare for the next scan.
14. Record the wavelengths for each color region of the spectra. Relate it to the color lines you observed through the diffraction grating. For each peak, move your cursor to the top of the peak. Click the left trackpad button. You should see the wavelength corresponding to the peak at the bottom of the screen. Make sure to mark down the wavelength (nm units).
15. Repeat steps 4 - 12 and record the emission spectra of all lamps, compiling data in your datasheet.

Experiment IV - Observation of light emission from commercial lighting

In part IV, you will investigate the emission spectrum from commercial light sources.

1. At your station you should have the following three lamps: white LED, colored LED and incandescent bulb. We will also test the room fluorescent ceiling lights. There is a fiber optic cable connected to a spectrometer.
2. On the computer screen, click on "Untitled" on the top left of the program (Fig. 10, A) and then select "New Experiment". The program will ask you if you want to save changes. Select "Don't Save".
3. A menu should appear. Select "Emissions" from the menu and then select "vs. Wavelength (Full Spectrum)". You are now ready to take your first scan.
4. Make sure the fume-hood light is turned off.
5. Turn on the white LED lamp. by pressing the push button on the neck of the lamp.
6. Aim the fiber optic cable at the lamp. The tip of the fiber optic should be roughly 1/2 meter away from the lamp. Press the "Collect" button on the software to begin data collection.
7. Once you see a spectrum appear on the screen press the "Stop" button.
8. On the right of the screen, you should see a data table. Next to (Fig. 10, C) "Data Set 1", click on the three dots and click on "Rename Data Set". Rename the data set based on the lamp that you are using. For example, name the dataset "White LED" if you are using the White LED.
9. If there are more than one spectrum displayed on the graph (this will be the case once you scan your second lamp) you can focus on the most recent scan by clicking on the y-axis label (Fig. 10, E) "Intensity" on the left side of the screen. Deselect "Intensity" for any but the most recent spectra. This will clear the data to allow us to prepare for the next scan.
10. Record the wavelengths of the 4 most intense peaks on your Data Sheet/Results. For any spectrum that is broad and continuous (not a discrete sharp peaks) mark down the range of the broad signal. For each peak, move your cursor to the top of the peak. Click the left trackpad button. You should see the (Fig. 10, D) wavelength corresponding to the peak at the bottom of the screen. You can drag a cursor around the signal to find the minimum and maximum wavelength. Make sure to mark down the wavelength (nm units).
11. Repeat steps 7 - 12 and record the emission spectra of all 3 lamps as well as the fluorescent light in the lab ceiling. Compile your data in your datasheet.

Extra Problems (Not for credit)

1. What wavelength corresponds to electro-magnetic radiation with a frequency of 1.9985×10^{14} Hz

Answers

2. What is the speed of an ocean wave with a wavelength of 4.5 m and a frequency of frequency of 3.5 waves per second?

Answers

3. Calculate the energy of the photon emitted when an electron in a hydrogen atom moves from $n=5$ to $n=1$.

Answers

4. In figure 2 in the lab manual, which graph best represents a monochromatic light source?
 - a. A
 - b. B
 - c. C
 - d. D

Answers

5. Using a spectrometer, you have determined that the wavelength of the photon emitted from a hydrogen discharge lamp is 93.2 nm. You know the electron transition occurred from $n=4$ to $n=1$. Determine the experimental Bohr constant based on this data.

Answers

6. Which chemical in the lab is a chemical irritant?
 - a. Lithium chloride
 - b. Sodium chloride
 - c. Copper chloride
 - d. Calcium chloride
 - e. All of the above

Answers